

Shruti Sriram

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EDUCATION

The University of Texas at Austin

Aug 2024 - May 2026

Master of Science in Computer Science

CGPA: 3.85/4.0

Focus Areas: Software Development, AI/ML, Computer Vision

Courses: Visual Recognition, Generative Models in ML, Spoken Language Technologies, Statistics, Database Systems, Distributed Computing

Sri Sivasubramaniya Nadar College of Engineering

Aug 2020 - Jun 2024

Bachelor of Engineering in Computer Science and Engineering (Gold Medal in Semester 1, Silver Medal in Semester 3)

CGPA: 9.61/10

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, HTML, CSS, JavaScript, Bash, Go, Rust

Tools & Frameworks: React, Node.js, Tableau, Apache Camel, Spring Boot

ML Stack: PyTorch, TensorFlow, Keras, OpenCV, Scikit-Learn, Numpy, Pandas

AI Frameworks and Agentic Systems: OpenAI API, Claude, Cursor, Ollama, LangChain, CrewAI

Databases: MySQL, PostgreSQL, DuckDB, MongoDB

Cloud & DevOps: AWS, Git

EXPERIENCE

The University of Texas at Austin, Graduate Researcher (System Level Design Group)

Oct 2025 - Present

- Designed a novel, computationally efficient medical image segmentation architecture for cross-domain settings, improving DICE by up to +18.9% on challenging domain-shifted datasets
- Developed an edge-optimized deployment pipeline for the model to reduce inference latency and memory footprint on resource-constrained devices
- Building a real time segmentation device for colorectal procedures, enabling high accuracy with low computation to generate instantaneous segmentation masks

Ascension, Inc., Engineering Intern

Jun 2025 - August 2025

- Built code analysis tools for the AAVA Edge platform, including Abstract Syntax Tree (AST) parsers (using pyparser, javalang, and esprima), visualizations, and call graphs, to support faster code understanding and refactoring
- Developed a DuckDB backend and Thespian actor abstraction to store parsed ASTs and AI agent outputs; this set-up enables multi-user access and versioning, improves data accessibility and reliability, thus enhancing code traceability and reproducibility
- Designed a NiceGUI dashboard with real-time drill-down views, filtering and querying features, to expedite engineer onboarding time by 40% and debugging time by 60%

Citibank, Summer Analyst Intern

May 2023 - Jul 2023

- Re-engineered legacy Ab Initio billing feed pipeline to automate periodic ingestion of 50K+ billing records per day using Apache Camel, Java, and Spring Boot
- Reduced manual intervention by 90% by scheduling recurring jobs using cron expressions
- Built a Tableau dashboard on geo-temporal data to help business analysts monitor trends, spot anomalies, and support forecasting and planning
- Performed hypothesis testing to identify temporal anomalies and helped stakeholders frame data-driven questions more effectively

Synergy Maritime, Software Development Intern

Aug 2022 - Aug 2022

- Designed an internal web app to manage ship details, including IMO numbers, vessel dimensions, and voyage records
- Developed a frontend using HTML/CSS and SQL Server-backed queries with integrated form validation, to optimize data retrieval time from 15 seconds to under 5 seconds
- Built dashboards using Pandas, NumPy, and Matplotlib to visualize 100K+ records from vessel sensors and operations, and aid anomaly detection

PUBLICATIONS

SACNet: Scale-Adaptive Convolutional Decoder for Generalizable Polyp Segmentation

Submitted to MICCAI 2026

Multimodal Forgery Detection in Videos Using a Tri-Network Model ([Paper Link](#))

IEEE Xplore

PoSh at SemEval-2023 Task 10: Explainable Detection of Online Sexism ([Paper Link](#))

ACL Anthology

A Fusion Approach for Web Search Result Diversification Using Machine Learning Algorithms ([Paper Link](#))

CEUR Workshop

PROJECTS

LLM Prompt Engineering for Data Cleaning (PHI-3)

Aug 2024 - Dec 2024

- Identified optimal prompt strategies by evaluating Phi-3's effectiveness in cleaning files with diverse error types, by benchmarking multiple prompt patterns to measure output quality

911 Emergency Response System Enhancement

Nov 2024 - Nov 2024

- Built a multi-agent AI system to augment 911 emergency dispatch, with specialized agents for threat analysis, multilingual translation, and multimodal input routing
- Each agent processes a distinct signal: caller tone, background audio, or video input, using LLaMA 3.2-Vision, Whisper, and LangChain
- Improved dispatcher prioritization and accessibility for non-English speakers and disabled users

Segmentation and 3D Reconstruction of Arteries, Istinye University, Turkey

Dec 2022 - Oct 2024

- Developed a pipeline for arterial segmentation and 3D reconstruction using UNet and TransUNet, integrating K-means clustering and edge detection to generate precise geometric models
- Achieved significantly higher segmentation fidelity with UNet (IoU 0.89 vs. 0.65), enabling more reliable anatomical modeling for downstream clinical analysis and surgical planning

Automatic Estimation of Fish Count in Aqua Farms

Jan 2022 - May 2024

- Built a YOLOv8- and ByteTrack-based fish counting and tracking system, using a Raspberry Pi and a webcam, for smart aquaculture and automated feed-dispensing